

Causal Phase Obstructions in Exact Information Simulation

Zero-Repair Trees, Hidden Ordering Information, and Multiscale Renormalization

Jan Mikulík

23 August 2026

Revised phase-fiber version 3

Abstract

We study an exact causal information-accounting problem in which two finite information-increment laws must be reconciled by a nonnegative, history-dependent inventory, with no fresh repair term. The local constraint is a linear convolution identity, but the global feasibility problem is strongly order-sensitive.

Our first result is a finite-horizon converse. For every fixed Laplace tilt we derive an explicit logarithmic lower bound for the minimum root inventory $\mathcal{B}_n$. If the increment laws have equal mean and variance and their first mismatch is an oriented third cumulant, then

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{\Delta \kappa_3}{6\sigma^2}.$$

For the dyadic pair

$$\mu_P = \tfrac{1}{4}\delta_2 + \tfrac{3}{4}\delta_4, \quad \mu_Q = \tfrac{3}{4}\delta_3 + \tfrac{1}{4}\delta_5,$$

this gives the exact lower-bound coefficient $1/3$ and rules out every infinite finite-mean zero-repair tree.

We then isolate a second obstruction which is invisible to scalar information spectra. At block length m , ordered binary histories form a space $V_m \cong \mathbb{R}^{2^m}$, while scalar total information sees only the $m+1$ Hamming layers. The lost ordering information is the phase space $W_m = \ker \Pi_m$, of dimension

$$\dim W_m = 2^m - (m+1).$$

Under block doubling it has the exact decomposition

$$W_{2m} = (W_m \otimes W_m) \oplus (W_m \otimes U_m) \oplus (U_m \otimes W_m) \oplus K_m, \quad \dim K_m = m^2,$$

where K_m is a fresh anti-diagonal phase sector created by coarse-graining. For two symbols we go further: the entire projected causal scalar-profile polytope is the static transport polytope cut by exactly two additional linear facets,

$$2r_{00} + 3r_{01} + r_{11} \geq \frac{3}{16}, \quad 2r_{00} + 3r_{01} + 2r_{10} + 3r_{11} \leq \frac{9}{8}.$$

This intrinsic criterion yields a solver-free proof that the minimum information-profile mismatch is $5/16$, whereas static matching pays only $1/4$; a fully displayed rational primal–dual certificate gives an independent exact check. More generally, at every fixed block length the local causal phase fiber is a bounded polytope. We give an exact necessary-and-sufficient Farkas characterization of its nonemptiness; for rational source laws the certificate cone is rational polyhedral and hence has a finite rational generating family. The extra $1/16$ in the two-symbol example is added to both scalar residual profiles and therefore disappears from their signed difference. Causality can thus create a real obstruction living entirely in the kernel of scalar information projection.

Finally, we pass from viability sets to lower support functionals, derive an exact finite-dimensional Bellman–Fenchel recursion, identify the cubic shift normal form

$$P^* - Q^* = \frac{1}{4}E^2(I - E)^3,$$

and introduce a Green coordinate in which the cubic defect becomes a positive unit source. The two-block boundary mask is $(1 + z)^3/4$, the binary quadratic B-spline mask, and one explicit mean-neutral phase channel admits a uniform finite-mean envelope across all dyadic scales. These statements do not prove a matching $O(\log n)$ upper bound. Fixed-block phase feasibility is now completely characterized at the level of finite-dimensional linear programming; what remains is the genuinely multiscale problem of finding a scale-stable description compatible with future zero-repair viability. In particular, a uniform bound on $\mathcal{B}_{2n} - \mathcal{B}_n$ would close the order of growth.

The zero-repair bill is deliberately distinguished from unrestricted causal Shannon entropy. It measures exact causal working inventory, not entropy rate; this distinction is the link back to the broader question of which resources an exact stochastic realization must carry.

Keywords: causal information, exact simulation, zero repair, causal transport, information spectrum, renormalization, Bellman–Fenchel duality, subdivision schemes, third cumulant.

1 Introduction

A probability law tells us what may be observed. It does not, by itself, tell us how much hidden state an exact realization must carry when outputs are chosen sequentially and future repair is forbidden. This paper isolates one such resource question.

Suppose two actions have prescribed information-increment laws μ_P and μ_Q . At every history we ask for an exact accounting identity: the current nonnegative inventory plus the next Q -increment must have the same law as a P -branch increment plus the next inventory assigned to that branch. No signed debt is allowed and no fresh repair term may be injected. The resulting object is an exact zero-repair information tree.

There are two reasons to study this stricter contract separately from ordinary entropy-rate simulation. First, exact online random generation can recycle unused randomness and operate arbitrarily close to Shannon entropy in models that allow a persistent state and fresh source bits [13, 14]. Second, finitary coding theory shows that equal entropy alone does not control stronger moments of coding radius or information cocycles [2, 3, 4]. Thus rate and working state are different resources.

The present problem also sits near, but not inside for free, modern adapted optimal transport. Causal and bicausal transport admit general duality and dual attainment under broad assumptions [9]; graph-causal transport now provides gluing and dynamic-programming results for couplings constrained by directed acyclic graphs [16]. Our model has an additional exact conservation law and a hard state constraint $J \geq 0$. These are not cosmetic: they are the source of the obstruction.

A useful recent methodological parallel comes from synchronization channels. Kovačević identifies a coefficientwise domination condition for a family of sticky channels, converts it into an exact dual capacity bound, and then characterizes the laws satisfying that condition through a functional composition equation [15]. The present setting is different, but the same standard is appropriate: the objective is not another successful ansatz, but an exact structural criterion exposing the admissible positive fiber on which the dynamics acts.

1.1 Main contributions

The paper has six main components.

1. We give a self-contained finite-horizon logarithmic converse for exact zero repair. The third-cumulant case yields an explicit lower-bound coefficient, equal to $1/3$ for the dyadic

parity pair.

2. We represent finite viability sets by their lower support functionals and derive an exact Bellman–Fenchel recursion in finite dimensions, together with a rigorous weak-duality version on the countable state space.
3. We introduce the ordered-history phase quotient. Scalar total information is a projection Π_m from 2^m ordered histories to $m + 1$ Hamming layers; its kernel has dimension $2^m - (m + 1)$.
4. We characterize the first nontrivial projected causal phase polytope in closed form. For two symbols a static scalar joint profile admits a causal lift if and only if two explicit linear facet inequalities hold. This gives a solver-free proof of the exact mismatch $5/16$, compared with the static value $1/4$, while an independent rational primal–dual certificate checks the same optimum. We then characterize the local causal phase fiber at arbitrary fixed block length as a polytope and give an exact Farkas dual criterion for its nonemptiness.
5. We prove a block-doubling decomposition of the phase kernel. Besides inherited phase directions, each doubling creates a fresh m^2 -dimensional anti-diagonal sector. This rules out any globally one-dimensional phase picture and shows why a fixed-block Farkas description is not yet a scale-uniform renormalization theorem.
6. We identify the cubic finite-difference normal form, a Green-coordinate affine recursion, the associated quadratic B-spline boundary mask, and one explicit marginal phase channel that can be transported through all dyadic scales at bounded mean cost. The sharp upper bound remains open; the paper ends with a precise renormalization target rather than a claimed QED.

1.2 Resource interpretation and scope

The quantity studied here is not automatically the excess Shannon entropy of an unrestricted causal simulator. It is an auxiliary working-inventory cost under a stricter contract. We will return to this distinction in Section 18. In particular, no ontological conclusion is drawn from the mathematics: the paper prices a stated realization constraint.

2 Exact zero-repair information trees

Let

$$\mu_P = \sum_{i=1}^r p_i \delta_{x_i}, \quad p_i > 0, \quad \sum_i p_i = 1,$$

and let μ_Q be a finite-support probability law on $[0, \infty)$. Branch labels are retained even if two values x_i coincide; equal current information does not imply equal future subtrees.

Definition 2.1 (Depth- n exact zero-repair tree). A depth- n zero-repair tree is a family of probability laws

$$\{B_h\}_{|h| \leq n}$$

on $[0, \infty)$, indexed by histories $h \in \{1, \dots, r\}^{\leq n}$, such that each internal node satisfies

$$\mu_Q * B_h = \sum_{i=1}^r p_i \delta_{x_i} * B_{hi}. \tag{1}$$

For a buffer law B_h , put $m_h := \mathbb{E}_{B_h} J$.

Definition 2.2 (Finite-horizon zero-repair bill).

$$\mathcal{B}_n := \inf \left\{ m_\emptyset : \{B_h\}_{|h| \leq n} \text{ satisfies (1)} \right\}. \tag{2}$$

The value is $+\infty$ if no feasible tree exists.

Throughout the converse we assume equal increment means,

$$\mathbb{E}_{\mu_P} X = \mathbb{E}_{\mu_Q} Y. \quad (3)$$

Lemma 2.3 (Mean martingale). *If $m_h < \infty$, then all child means are finite and*

$$\boxed{m_h = \sum_i p_i m_{hi}.} \quad (4)$$

Consequently, along an independently P -distributed branch history, m_{H_t} is a nonnegative martingale.

Proof. Take first moments in (1). Finite support and nonnegative child means imply finiteness of each child mean. The common increment mean cancels, leaving (4). $\square$

Remark 2.4 (Constrained coboundary form). After coupling the equal laws in (1), one may write schematically

$$J_h + Y = x_I + J_{hI}.$$

Thus $Y - x_I = J_{hI} - J_h$ resembles a coboundary. The difficulty is the simultaneous imposition of nonnegativity, exact branch weights, convolutional input, and future viability at every history.

3 A finite-horizon logarithmic converse

Fix $a > 0$ and define

$$G_P(a) := \mathbb{E}_{\mu_P} e^{-aX}, \quad G_Q(a) := \mathbb{E}_{\mu_Q} e^{-aY}, \quad \rho(a) := \frac{G_Q(a)}{G_P(a)}.$$

Introduce the tilted branch law

$$\tilde{p}_i^{(a)} := \frac{p_i e^{-ax_i}}{G_P(a)}$$

and its likelihood-ratio second moment

$$\boxed{\Lambda(a) := \sum_i \frac{(\tilde{p}_i^{(a)})^2}{p_i} = \frac{G_P(2a)}{G_P(a)^2} \geq 1.} \quad (5)$$

For each node set

$$\hat{B}_h(a) := \mathbb{E}_{B_h} e^{-aJ}, \quad y_h := e^{-am_h}.$$

By Jensen and $J \geq 0$,

$$0 < y_h \leq \hat{B}_h(a) \leq 1. \quad (6)$$

For a k -step descendant hw , define

$$D_h^{(k)} := \mathbb{E}_{P^k} y_{hw} - y_h.$$

Lemma 3.1 (Jensen–Hellinger control). *For every node and every admissible k ,*

$$D_h^{(k)} \geq 0,$$

and

$$\mathbb{E}_{P^k} (\sqrt{y_{hw}} - \sqrt{y_h})^2 \leq D_h^{(k)}, \quad \mathbb{E}_{P^k} (y_{hw} - y_h)^2 \leq 4D_h^{(k)}. \quad (7)$$

Proof. By (4), $\mathbb{E}_{P^k} m_{hw} = m_h$. Convexity of $u \mapsto e^{-au}$ gives $D_h^{(k)} \geq 0$, while

$$\mathbb{E} e^{-am_{hw}/2} \geq e^{-a\mathbb{E} m_{hw}/2} = \sqrt{y_h}.$$

Expanding the square proves the first estimate. Since $|u - v| \leq 2|\sqrt{u} - \sqrt{v}|$ on $[0, 1]$, the second follows. $\square$

Taking Laplace transforms in (1) yields the exact tilted identity

$$\boxed{\mathbb{E}_{\tilde{P}_a} \hat{B}_{hi}(a) = \rho(a) \hat{B}_h(a), \quad \mathbb{E}_{\tilde{P}_a^k} \hat{B}_{hw}(a) = \rho(a)^k \hat{B}_h(a).} \quad (8)$$

Let

$$L_w := \frac{\tilde{P}_a^k(w)}{P^k(w)}.$$

Then $\mathbb{E} L_w = 1$ and $\mathbb{E} L_w^2 = \Lambda(a)^k$.

Lemma 3.2 (One-block measure change). *For every node h ,*

$$\boxed{\rho(a)^k \hat{B}_h(a) \geq y_h - 2\Lambda(a)^{k/2} \sqrt{D_h^{(k)}}.} \quad (9)$$

Proof. Since $\mathbb{E}(L_w - 1) = 0$,

$$\mathbb{E}_{\tilde{P}_a^k} y_{hw} = y_h + \mathbb{E}_{P^k} [(L_w - 1)(y_{hw} - y_h)].$$

Cauchy-Schwarz and (7) give the stated lower bound. Finally use $\hat{B}_{hw}(a) \geq y_{hw}$ and (8). $\square$

Proposition 3.3 (Master inequality). *Let a feasible depth- n tree have root mean m . For every integer $k \leq n/2$,*

$$\boxed{\rho(a)^k \geq e^{-am} - 2\Lambda(a)^{k/2} \sqrt{\frac{2amk}{n}}.} \quad (10)$$

Proof. Put $q = \lfloor n/k \rfloor \geq n/(2k)$. The disjoint k -blocks telescope:

$$\sum_{j=0}^{q-1} \mathbb{E} D_{H_{jk}}^{(k)} = \mathbb{E} y_{H_{qk}} - y_{\emptyset} \leq 1 - e^{-am} \leq am.$$

Hence one block start satisfies

$$\mathbb{E} D_{H_{jk}}^{(k)} \leq \frac{2amk}{n}.$$

Average (9) over that starting history, use $\mathbb{E} m_{H_{jk}} = m$, Jensen, $\hat{B}_h \leq 1$, and $\mathbb{E} \sqrt{D} \leq \sqrt{\mathbb{E} D}$. $\square$

Theorem 3.4 (Fixed-tilt logarithmic obstruction). *Assume (3). If $\rho(a) < 1$ for some fixed $a > 0$, set*

$$d(a) := -\log \rho(a) > 0, \quad \Gamma(a) := 2a + \frac{a \log \Lambda(a)}{d(a)}.$$

Then

$$\boxed{\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{1}{\Gamma(a)}} \quad (11)$$

Here and below $\log$ denotes the natural logarithm.

Proof. For a feasible depth- n tree of root mean m , choose

$$k = \left\lceil \frac{am + \log 2}{d(a)} \right\rceil.$$

Then $\rho(a)^k \leq \frac{1}{2}e^{-am}$. If $k > n/2$, the definition of k already gives a lower bound on m linear in n , stronger than (11). Otherwise combine this estimate with (10) and square to obtain

$$n \leq 32amk \Lambda(a)^k e^{2am}.$$

The ceiling bound on k therefore yields a finite constant C_a such that

$$n \leq C_a(m+1)^2 e^{\Gamma(a)m}.$$

Taking logarithms,

$$\log n \leq \Gamma(a)m + 2\log(m+1) + \log C_a.$$

Apply this inequality to a sequence of feasible trees whose root means are within $1/n$ of $\mathcal{B}_n$, and pass to a subsequence realizing the liminf. If the ratio is finite, then $m = O(\log n)$ on that subsequence and $\log(m+1) = o(\log n)$. Division by $\log n$ proves (11). $\square$

Theorem 3.5 (Third-cumulant logarithmic obstruction). *Assume that μ_P and μ_Q have common mean μ , common variance $\sigma^2 > 0$, and*

$$\Delta\kappa_3 := \kappa_3(\mu_Q) - \kappa_3(\mu_P) > 0.$$

Then

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{\Delta\kappa_3}{6\sigma^2}. \quad (12)$$

In particular $\mathcal{B}_n \rightarrow \infty$.

Proof. Let

$$K_P(a) := \log \mathbb{E}e^{-aX}, \quad K_Q(a) := \log \mathbb{E}e^{-aY}.$$

Finite support gives, as $a \downarrow 0$,

$$K_P(a) = -\mu a + \frac{\sigma^2}{2}a^2 - \frac{\kappa_3(P)}{6}a^3 + O(a^4),$$

$$K_Q(a) = -\mu a + \frac{\sigma^2}{2}a^2 - \frac{\kappa_3(Q)}{6}a^3 + O(a^4).$$

Hence

$$d(a) = K_P(a) - K_Q(a) = \frac{\Delta\kappa_3}{6}a^3 + O(a^4),$$

and

$$\log \Lambda(a) = K_P(2a) - 2K_P(a) = \sigma^2 a^2 + O(a^3).$$

Therefore

$$\Gamma(a) \longrightarrow \frac{6\sigma^2}{\Delta\kappa_3}.$$

Apply Theorem 3.4 for each sufficiently small fixed $a > 0$, first take $n \rightarrow \infty$, and only then send $a \downarrow 0$. $\square$

4 The dyadic parity pair

We now specialize to

$$\boxed{\mu_P = \frac{1}{4}\delta_2 + \frac{3}{4}\delta_4, \quad \mu_Q = \frac{3}{4}\delta_3 + \frac{1}{4}\delta_5.} \quad (13)$$

Both laws have mean $7/2$, variance $3/4$, and centered third cumulants

$$\kappa_3(P) = -\frac{3}{4}, \quad \kappa_3(Q) = +\frac{3}{4}.$$

Thus $\Delta\kappa_3 = 3/2$.

Corollary 4.1 (The $1/3$ lower law). *For (13),*

$$\boxed{\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \frac{1}{3}.} \quad (14)$$

Equivalently,

$$\mathcal{B}_n \geq \left(\frac{1}{3} - o(1)\right) \log n.$$

For base-two logarithms the coefficient is $(\log 2)/3$.

With $z = e^{-a}$, the increment PGFs are

$$G_P(z) = \frac{1}{4}z^2 + \frac{3}{4}z^4, \quad G_Q(z) = \frac{3}{4}z^3 + \frac{1}{4}z^5,$$

and

$$\boxed{G_P(z) - G_Q(z) = \frac{1}{4}z^2(1-z)^3.} \quad (15)$$

Corollary 4.2 (No finite-mean infinite zero-repair tree). *There is no infinite history-indexed zero-repair tree for (13) with finite root mean.*

Proof. Such a tree would restrict to a feasible depth- n tree for every n , forcing $\mathcal{B}_n$ to be uniformly bounded by its root mean, contrary to (14). $\square$

5 Viability kernels and Bellman–Fenchel duality

For the dyadic pair, write the probability generating function of a buffer law as the same symbol. Multiplying (1) by four gives

$$\boxed{B_{h0}(z) + 3z^2 B_{h1}(z) = A(z)B_h(z), \quad A(z) := z(3 + z^2).} \quad (16)$$

Define

$$\mathcal{K}_0 := \mathcal{P}(\mathbb{Z}_{\geq 0})$$

and recursively

$$\mathcal{K}_{n+1} := \{B \in \mathcal{P} : \exists B_0, B_1 \in \mathcal{K}_n \text{ satisfying (16)}\}. \quad (17)$$

Then

$$\mathcal{B}_n = \inf_{B \in \mathcal{K}_n} \mathbb{E}_B J.$$

Proposition 5.1 (Convexity). *Every $\mathcal{K}_n$ is convex.*

Proof. The base $\mathcal{K}_0$ is convex. If $B, B' \in \mathcal{K}_{n+1}$ have feasible child pairs (B_0, B_1) and (B'_0, B'_1) , then linearity of (16) shows that every convex combination of B, B' has the corresponding convex combinations of the children, which lie in $\mathcal{K}_n$. $\square$

For a convex set K of probability laws and a test function j , define its lower support functional by

$$\ell_K(j) := \inf_{B \in K} \langle j, B \rangle. \quad (18)$$

Write $\ell_n := \ell_{\mathcal{K}_n}$. Then

$$\mathcal{B}_n = \ell_n(\text{id}), \quad \text{id}(t) = t.$$

On a finite state space, lower support is a complete coordinate for a closed convex viability set.

Proposition 5.2 (Lossless finite-dimensional support representation). *Let X be finite, Δ_X the probability simplex, and $K \subseteq \Delta_X$ nonempty, closed and convex. Then*

$$K = \bigcap_{j \in \mathbb{R}^X} \{b \in \Delta_X : \langle j, b \rangle \geq \ell_K(j)\}. \quad (19)$$

Proof. The forward inclusion is immediate. If $b \notin K$, finite-dimensional separation gives j such that $\langle j, b \rangle < \inf_{x \in K} \langle j, x \rangle$. $\square$

We now state the abstract dual recursion. Let X, Y be finite sets, $K \subseteq \Delta_X$ nonempty compact convex, and $C, S_1, \dots, S_r : \mathbb{R}^X \rightarrow \mathbb{R}^Y$ linear. For positive weights p_i with $\sum_i p_i = 1$, define

$$\Phi(K) := \left\{ b \in \Delta_X : \exists b_i \in K, \quad Cb = \sum_i p_i S_i b_i \right\}.$$

Theorem 5.3 (Bellman–Fenchel duality). *If $\Phi(K) \neq \emptyset$, then for every $j \in \mathbb{R}^X$,*

$$\ell_{\Phi(K)}(j) = \sup_{\lambda \in \mathbb{R}^Y} \left\{ \min_{x \in X} (j_x - (C^* \lambda)_x) + \sum_i p_i \ell_K(S_i^* \lambda) \right\}. \quad (20)$$

Proof. Consider the convex program minimizing $\langle j, b \rangle$ over $(b, b_1, \dots, b_r) \in \Delta_X \times K^r$ subject to the displayed linear equality. With multiplier λ , the Lagrangian is

$$\langle j - C^* \lambda, b \rangle + \sum_i p_i \langle S_i^* \lambda, b_i \rangle.$$

Minimizing over the product set gives the right-hand side of (20). The primal feasible set is compact and convex and the constraints are linear, so finite-dimensional strong duality follows from separation [1]. $\square$

For the dyadic model,

$$\begin{aligned} (C_Q^* \lambda)(t) &= \frac{3}{4} \lambda(t+3) + \frac{1}{4} \lambda(t+5), \\ (S_2^* \lambda)(t) &= \lambda(t+2), \quad (S_4^* \lambda)(t) = \lambda(t+4). \end{aligned}$$

Thus the formal countable-state Bellman transform is

$$\begin{aligned} (\mathfrak{D}\ell)(j) = \sup_{\lambda} \left\{ \inf_{t \geq 0} \left[j(t) - \frac{3}{4} \lambda(t+3) - \frac{1}{4} \lambda(t+5) \right] \right. \\ \left. + \frac{1}{4} \ell(\lambda(\cdot+2)) + \frac{3}{4} \ell(\lambda(\cdot+4)) \right\}. \end{aligned} \quad (21)$$

On a finite exact truncation, $\ell_{n+1} = \mathfrak{D}\ell_n$. On the full countable state space, the same expression is always a valid lower certificate for each admissible λ , but equality requires a separate no-gap theorem in a suitable weighted function space.

Proposition 5.4 (Weak countable-state recursion). *For every admissible λ for which the pairings are finite,*

$$\begin{aligned} \ell_{n+1}(j) \geq \inf_{t \geq 0} \left[j(t) - \frac{3}{4}\lambda(t+3) - \frac{1}{4}\lambda(t+5) \right] \\ + \frac{1}{4}\ell_n(\lambda(\cdot+2)) + \frac{3}{4}\ell_n(\lambda(\cdot+4)). \end{aligned} \quad (22)$$

Proof. For any feasible parent and children, add the zero pairing of λ with the constraint. Bound the parent contribution by its pointwise infimum and each child contribution by the corresponding lower support, then take the infimum over feasible triples. $\square$

The operator is order preserving and additively homogeneous on every domain where the expressions are finite:

$$\ell_1 \leq \ell_2 \Rightarrow \mathfrak{D}\ell_1 \leq \mathfrak{D}\ell_2, \quad \mathfrak{D}(\ell + c) = \mathfrak{D}\ell + c.$$

Consequently, on any invariant class on which the uniform norm is finite,

$$\|\mathfrak{D}\ell_1 - \mathfrak{D}\ell_2\|_\infty \leq \|\ell_1 - \ell_2\|_\infty.$$

There is also the exact multiplier gauge $\lambda \sim \lambda + c$, a direct consequence of probability normalization. These are the structural features of a topical or Shapley-type operator; they motivate, but do not by themselves justify, nonlinear Perron–Frobenius methods [5, 8, 7].

6 Ordered histories and the phase quotient

The scalar information spectrum forgets order. The causal tree does not. We now make the difference algebraic.

Let

$$V_m := \mathbb{R}^{\{0,1\}^m}$$

with canonical basis e_w , $w \in \{0,1\}^m$. For a binary word w , let $\text{wt}(w)$ denote its Hamming weight. Define the scalarization map

$$\boxed{\Pi_m : V_m \rightarrow \mathbb{R}^{m+1}, \quad \Pi_m e_w = e_{\text{wt}(w)}}. \quad (23)$$

For a signed measure on ordered histories, Π_m simply sums its masses within equal-Hamming-weight layers. Since the total information of either binary source is an affine function of Hamming weight, this is exactly the projection from ordered histories to scalar total-information profiles.

Definition 6.1 (Ambient phase space).

$$\boxed{W_m := \ker \Pi_m}. \quad (24)$$

Elements of W_m are signed rearrangements invisible to scalar total information.

Theorem 6.2 (Dimension of hidden ordering information). *For every $m \geq 1$,*

$$\boxed{\dim W_m = 2^m - (m+1)}. \quad (25)$$

Proof. The map Π_m is surjective: for every $0 \leq k \leq m$, any word of weight k maps to the k -th basis vector. Hence $\text{rank } \Pi_m = m+1$, and rank-nullity gives (25). $\square$

Define normalized layer vectors

$$u_k^{(m)} := \binom{m}{k}^{-1} \sum_{\text{wt}(w)=k} e_w, \quad 0 \leq k \leq m,$$

and let

$$U_m := \text{span}\{u_0^{(m)}, \dots, u_m^{(m)}\}.$$

Then $\Pi_m|_{U_m}$ is an isomorphism and

$$V_m = U_m \oplus W_m. \quad (26)$$

Thus U_m is a canonical visible sector and W_m the invisible phase sector.

For $m = 2$,

$$u_0 = e_{00}, \quad u_1 = \frac{1}{2}(e_{01} + e_{10}), \quad u_2 = e_{11},$$

and

$$W_2 = \text{span}\{\theta\}, \quad \theta := \frac{1}{2}(e_{01} - e_{10}). \quad (27)$$

This is the first nontrivial ordering phase: the words 01 and 10 have the same scalar information but different causal pasts.

6.1 Two-block phase coordinates for buffer PGFs

Iterating (16) twice gives

$$\boxed{B_{00} + qB_{01} + qB_{10} + q^2B_{11} = A^2B, \quad q := 3z^2.} \quad (28)$$

Put

$$S := \frac{B_{01} + B_{10}}{2}, \quad \Theta := \frac{B_{01} - B_{10}}{2}. \quad (29)$$

Then the coarse equation loses Θ :

$$\boxed{B_{00} + 2qS + q^2B_{11} = A^2B.} \quad (30)$$

Proposition 6.3 (Exact two-block lifting criterion). *Let B, B_{00}, S, B_{11} be probability PGFs satisfying (30). They admit a genuine two-step zero-repair lift if and only if there exists a signed series Θ with $\Theta(1) = 0$ such that*

$$B_{01} = S + \Theta, \quad B_{10} = S - \Theta$$

are probability PGFs and

$$\boxed{B_0 = \frac{B_{00} + q(S + \Theta)}{A}, \quad B_1 = \frac{S - \Theta + qB_{11}}{A}} \quad (31)$$

are probability PGFs.

Proof. Necessity follows by taking S, Θ from an existing two-step tree and solving the first-level recursions for B_0, B_1 . Conversely, if the four PGFs in the statement are nonnegative and normalized, the formulas imply

$$B_{00} + qB_{01} = AB_0, \quad B_{10} + qB_{11} = AB_1,$$

and adding $B_0 + qB_1 = AB$ is equivalent to (30). Hence the full two-step tree closes exactly. $\square$

Remark 6.4. Proposition 6.3 is a change of coordinates, not a solution of the positivity problem. Its value is that it isolates the missing datum: the coarse scalar equation determines the visible part, while causal feasibility asks whether the corresponding phase fiber contains a point compatible with all PGF inequalities.

7 Exact two-symbol causal phase geometry

The phase quotient is not merely formal. At the first nontrivial block length one can characterize *exactly* which scalar joint profiles admit an ordered causal lift. This gives a closed-form phase-feasibility criterion, not merely an LP description.

Let $X = (X_1, X_2) \sim P^{\otimes 2}$ and $W = (W_1, W_2) \sim Q^{\otimes 2}$, where

$$P(0) = \frac{1}{4}, \quad P(1) = \frac{3}{4}, \quad Q(0) = \frac{3}{4}, \quad Q(1) = \frac{1}{4}.$$

For a coupling π , define its scalar Hamming-layer profile

$$r_{ij} := \mathbb{P}_\pi\{\text{wt}(X) = i, \text{wt}(W) = j\}, \quad 0 \leq i, j \leq 2. \quad (32)$$

Its row and column marginals are fixed:

$$p^{(2)} = \frac{1}{16}(1, 6, 9), \quad q^{(2)} = \frac{1}{16}(9, 6, 1).$$

Write

$$a := r_{00}, \quad b := r_{01}, \quad c := r_{10}, \quad d := r_{11}, \quad S := a + b + c + d. \quad (33)$$

Then every static scalar coupling is uniquely of the form

$$r(a, b, c, d) = \begin{pmatrix} a & b & \frac{1}{16} - a - b \\ c & d & \frac{3}{8} - c - d \\ \frac{9}{16} - a - c & \frac{3}{8} - b - d & S - \frac{3}{8} \end{pmatrix}. \quad (34)$$

Thus static feasibility is exactly the nonnegativity of the nine entries in (34).

A coupling is causal from W to X if the conditional law of X_1 given (W_1, W_2) depends only on W_1 . Equivalently,

$$\sum_{x_2} \pi((a, x_2), (b, c)) = Q(c) \sum_{x_2, w_2} \pi((a, x_2), (b, w_2)) \quad (35)$$

for all $a, b, c \in \{0, 1\}$.

Theorem 7.1 (Exact two-symbol causal scalar-profile criterion). *Let $r = r(a, b, c, d)$ be a static scalar coupling with the fixed marginals above. Then r admits a causal lift π if and only if*

$$2a + 3b + d \geq \frac{3}{16}, \quad (36)$$

$$2a + 3b + 2c + 3d \leq \frac{9}{8}. \quad (37)$$

Equivalently, the projected causal polytope is obtained from the static transport polytope by intersecting it with precisely these two additional half-spaces.

More constructively, a causal lift exists if and only if the interval

$$L(a, b, c, d) \leq t \leq U(a, b, c, d) \quad (38)$$

is nonempty, where

$$L := \max \left\{ 0, \frac{4a}{3}, 4 \left(S - \frac{3}{8} \right), \frac{3}{8} - 2(b + d) \right\}, \quad (39)$$

$$U := \min \left\{ \frac{1}{4}, 4(a + b), \frac{3}{8} - 2b, \frac{4(a + c)}{3} \right\}. \quad (40)$$

Proof. Let

$$\alpha_{uv} := \mathbb{P}_\pi(X_1 = u, W_1 = v).$$

The one-symbol marginals force every such first-step coupling to have the form

$$\alpha(t) = \begin{pmatrix} t & \frac{1}{4} - t \\ \frac{3}{4} - t & t \end{pmatrix}, \quad 0 \leq t \leq \frac{1}{4}. \quad (41)$$

By causality and the product law of W ,

$$\mathbb{P}_\pi(X_1 = u, W_1 = v, W_2 = w) = Q(w)\alpha_{uv}(t). \quad (42)$$

For $u, v, w \in \{0, 1\}$, put

$$s_{uvw} := \pi((u, 0), (v, w)).$$

The companion mass with $X_2 = 1$ is then

$$\pi((u, 1), (v, w)) = Q(w)\alpha_{uv}(t) - s_{uvw}. \quad (43)$$

Hence a causal coupling is equivalent to choosing the eight numbers s_{uvw} inside their elementary capacity intervals.

The scalar profile determines the following quantities:

$$s_{000} = a, \quad s_{001} + s_{010} = b, \quad s_{011} = \frac{1}{16} - a - b, \quad (44)$$

$$s_{100} = a + c - \frac{3}{4}t, \quad s_{101} + s_{110} = b + d - \frac{3}{16} + \frac{1}{2}t, \quad s_{111} = \frac{3}{8} - S + \frac{1}{4}t. \quad (45)$$

Their capacity bounds from (42) are

$$\begin{aligned} 0 \leq s_{000} \leq \frac{3}{4}t, & \quad 0 \leq s_{001} \leq \frac{1}{4}t, & \quad 0 \leq s_{010} \leq \frac{3}{16} - \frac{3}{4}t, & \quad 0 \leq s_{011} \leq \frac{1}{16} - \frac{1}{4}t, \\ 0 \leq s_{100} \leq \frac{9}{16} - \frac{3}{4}t, & \quad 0 \leq s_{101} \leq \frac{3}{16} - \frac{1}{4}t, & \quad 0 \leq s_{110} \leq \frac{3}{4}t, & \quad 0 \leq s_{111} \leq \frac{1}{4}t. \end{aligned}$$

For two nonnegative bins of capacities u, v , a prescribed total R can be split between them if and only if $0 \leq R \leq u + v$. Applying this observation to the two sums in (44)–(45), and using static feasibility for the bounds independent of t , gives exactly (38)–(40).

It remains to eliminate t . Write the four lower bounds in L as L_0, L_1, L_2, L_3 in the displayed order, and similarly U_0, U_1, U_2, U_3 . Nonemptiness is equivalent to $L_i \leq U_j$ for every i, j . Under the static inequalities from (34), the only new comparisons are

$$L_3 \leq U_1 \iff 2a + 3b + d \geq \frac{3}{16}, \quad (46)$$

and

$$L_2 \leq U_3 \iff 2a + 3b + 2c + 3d \leq \frac{9}{8}. \quad (47)$$

For completeness, the remaining potentially nontrivial comparisons follow as follows. From $a + b \leq 1/16$ and (36), $b + d \geq 1/16$, hence $L_3 \leq U_0$. Also $S \leq 1/16 + 3/8 = 7/16$, which gives $L_2 \leq U_0$; the inequality $L_2 \leq U_1$ is just $c + d \leq 3/8$; and $L_2 \leq U_2$ follows from $S \leq 7/16$ and $b \leq 1/16$. Finally, $L_3 \leq U_3$ follows from $S \geq 3/8$, while comparisons involving L_0, L_1 , together with $L_3 \leq U_2$, are immediate from the static nonnegativity constraints. This proves that (36)–(37) are necessary and sufficient.

For constructive sufficiency choose any $t \in [L, U]$. Set the four singleton quantities in (44)–(45). Split b between s_{001}, s_{010} within their two capacities, and split $b + d - 3/16 + t/2$ between s_{101}, s_{110} within theirs. The interval conditions guarantee that both splits exist. Define every remaining cell by (43). All sixteen masses are nonnegative and obey (35). Equation (42) gives $W \sim Q^{\otimes 2}$. The matrix $\alpha(t)$ fixes the first-symbol marginal of X to be P ; together with the Hamming-weight row marginals $p^{(2)}$, this forces the ordered word probabilities $(1, 3, 3, 9)/16$, hence $X \sim P^{\otimes 2}$. By construction the coupling projects to r . Thus it is the desired causal lift. $\square$

Both facets are nonredundant. The static profile with $(a, b, c, d) = (0, 0, 3/8, 0)$ violates (36) while satisfying (37); the static profile with $(a, b, c, d) = (1/16, 0, 0, 3/8)$ satisfies (36) but violates (37).

Remark 7.2 (A first intrinsic phase criterion). Theorem 7.1 is stronger than saying that a finite LP can be solved. At the first nontrivial block it eliminates the hidden ordering variables completely: a static scalar profile has an ordered causal lift if and only if two explicit oblique facets are satisfied. The phase variable t is recoverable from the interval (38). This is the finite-block analogue of the kind of intrinsic positivity criterion one would want to propagate under renormalization.

The two-symbol total-information levels are

$$I_P(x) = 4 + 2 \text{wt}(x), \quad I_Q(w) = 6 + 2 \text{wt}(w).$$

Thus equal information occurs exactly for layer pairs $(i, j) = (1, 0)$ and $(2, 1)$. The matched mass of a scalar profile (34) is

$$M_{\text{match}}(r) = r_{10} + r_{21} = c + \frac{3}{8} - b - d, \quad (48)$$

so its mismatch is

$$M_{\text{mis}}(r) = \frac{5}{8} + b + d - c. \quad (49)$$

Corollary 7.3 (Exact two-symbol causal mismatch). *Among all causal couplings π with the prescribed product marginals,*

$$\boxed{\min_{\pi} \mathbb{P}_{\pi}\{I_P(X) \neq I_Q(W)\} = \frac{5}{16}.} \quad (50)$$

Equivalently, the maximum causally matchable equal-information mass is $11/16$. Without causality the minimum mismatch is $1/4$, so the causal surcharge is exactly $1/16$.

Proof. For every causally liftable scalar profile, (36) and $a + b \leq 1/16$ imply

$$\frac{3}{16} \leq 2a + 3b + d \leq \frac{1}{8} + b + d,$$

hence

$$b + d \geq \frac{1}{16}. \quad (51)$$

Static feasibility also gives $c + d \leq 3/8$. Since $d \geq 0$,

$$c - b - d = (c + d) - (b + 2d) \leq \frac{3}{8} - (b + d) \leq \frac{5}{16}.$$

Substitution into (49) yields

$$M_{\text{mis}}(r) \geq \frac{5}{8} - \frac{5}{16} = \frac{5}{16}.$$

Equality is attained at

$$(a, b, c, d) = \left(0, \frac{1}{16}, \frac{3}{8}, 0\right), \quad r = \frac{1}{16} \begin{pmatrix} 0 & 1 & 0 \\ 6 & 0 & 0 \\ 3 & 5 & 1 \end{pmatrix}. \quad (52)$$

This profile satisfies both (36) and (37), so Theorem 7.1 supplies a causal lift. Therefore the lower bound is sharp. $\square$

Remark 7.4 (Which facet prices the surcharge). The second causal facet (37) is genuine for the full projected profile geometry, but it is inactive at the optimizer of the mismatch functional. The exact $1/16$ surcharge is already forced by the first phase facet (36). In this precise sense the local causal price is a shadow of one hidden-ordering feasibility constraint.

7.1 Independent rational primal–dual certificate

The closed-form proof above does not require a solver. As an independent check, the original sixteen-variable causal LP admits exact rational primal and dual certificates.

Order rows and columns as 00, 01, 10, 11. The coupling

$$\pi = \frac{1}{16} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 3 & 2 & 3 & 1 \end{pmatrix} \quad (53)$$

has the required marginals, satisfies (35), projects to (52), and has mismatch 5/16.

Vectorize π row by row and write all four X -marginals, four W -marginals, and eight causal equations as $A\pi = b$, where

$$b = \frac{1}{16}(1, 3, 3, 9, 9, 3, 3, 1, 0, 0, 0, 0, 0, 0, 0, 0)^T. \quad (54)$$

The dual vector

$$y = \left(1, 0, -\frac{1}{4}, \frac{3}{4}; 0, 0, 0, -2; 0, 0, 0, 0, 1, 0, -3, 0\right) \quad (55)$$

satisfies $b^T y = 5/16$ and

$$A^T y = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 0 & 0 & -2 \\ 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \leq C := \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

entrywise. Hence weak duality independently certifies the same optimum. The complete rational matrix A and the multiplication are displayed in Appendix A.

The scalar two-symbol laws are

$$\mu_P^{*2} = \frac{1}{16}\delta_4 + \frac{6}{16}\delta_6 + \frac{9}{16}\delta_8, \quad (56)$$

$$\mu_Q^{*2} = \frac{9}{16}\delta_6 + \frac{6}{16}\delta_8 + \frac{1}{16}\delta_{10}. \quad (57)$$

Static matching pairs 6/16 at level 6 and 6/16 at level 8, for common mass 12/16. Hence

$$R_P^{\text{stat}} = \frac{1}{16}\delta_4 + \frac{3}{16}\delta_8, \quad R_Q^{\text{stat}} = \frac{3}{16}\delta_6 + \frac{1}{16}\delta_{10}.$$

The causal optimizer (53) matches 6/16 at level 6 but only 5/16 at level 8, so

$$R_P^{\text{causal}} = \frac{1}{16}\delta_4 + \frac{4}{16}\delta_8, \\ R_Q^{\text{causal}} = \frac{3}{16}\delta_6 + \frac{1}{16}\delta_8 + \frac{1}{16}\delta_{10}.$$

Consequently,

$$R_P^{\text{causal}} = R_P^{\text{stat}} + \frac{1}{16}\delta_8, \quad R_Q^{\text{causal}} = R_Q^{\text{stat}} + \frac{1}{16}\delta_8. \quad (58)$$

and therefore

$$R_P^{\text{causal}} - R_Q^{\text{causal}} = R_P^{\text{stat}} - R_Q^{\text{stat}}. \quad (59)$$

Corollary 7.5 (A causal obstruction invisible to signed scalar spectra). *The additional causal mismatch 1/16 in Corollary 7.3 leaves the signed scalar residual profile unchanged. Hence the signed information-spectrum difference alone cannot detect all causal matching obstructions.*

The first nontrivial causal phase obstruction is therefore fully visible in three equivalent forms: an ordered-history feasibility problem, the two intrinsic scalar facets (36)–(37), and an exact primal–dual LP certificate.

8 Phase creation under block doubling

The one-dimensional phase $e_{01} - e_{10}$ is special to two symbols. At larger scales the hidden sector grows rapidly, and block doubling creates new phase directions even if the input blocks are individually scalarized.

Identify

$$V_{2m} \cong V_m \otimes V_m$$

by concatenating two length- m words. Define

$$C_m : \mathbb{R}^{m+1} \otimes \mathbb{R}^{m+1} \rightarrow \mathbb{R}^{2m+1}$$

on basis vectors by

$$C_m(e_i \otimes e_j) = e_{i+j}. \quad (60)$$

Then the scalarization maps satisfy the exact factorization

$$\boxed{\Pi_{2m} = C_m \circ (\Pi_m \otimes \Pi_m)}. \quad (61)$$

Let

$$K_m := \ker \left(C_m|_{\Pi_m(U_m) \otimes \Pi_m(U_m)} \right), \quad (62)$$

and identify this kernel back inside $U_m \otimes U_m$ using the isomorphism $\Pi_m|_{U_m}$.

Theorem 8.1 (Exact phase decomposition under doubling). *For every $m \geq 1$,*

$$\boxed{W_{2m} = (W_m \otimes W_m) \oplus (W_m \otimes U_m) \oplus (U_m \otimes W_m) \oplus K_m}. \quad (63)$$

Moreover,

$$\boxed{\dim K_m = m^2}. \quad (64)$$

Proof. From (26),

$$V_m \otimes V_m = (U_m \otimes U_m) \oplus (U_m \otimes W_m) \oplus (W_m \otimes U_m) \oplus (W_m \otimes W_m).$$

Equation (61) annihilates every tensor component containing a factor from W_m . On $U_m \otimes U_m$, it is identified with C_m . Therefore the kernel is exactly the direct sum in (63).

The map C_m is surjective because every basis vector e_k , $0 \leq k \leq 2m$, equals $C_m(e_i \otimes e_j)$ for some $0 \leq i, j \leq m$ with $i + j = k$. Hence

$$\dim K_m = (m+1)^2 - (2m+1) = m^2.$$

□

The sector K_m has a simple interpretation. Even if each half-block has already been reduced to its Hamming layer, the total layer $i + j$ forgets how the weight was split between the two halves. Thus K_m is a genuinely *fresh anti-diagonal phase sector*: it is created by coarse-graining itself, not inherited from unresolved phase inside either sub-block.

8.1 The first doubling: two symbols to four

For $m = 2$, $\dim W_2 = 1$ and $\dim U_2 = 3$. Hence

$$\dim W_4 = 1 + 3 + 3 + 4 = 11,$$

in agreement with $2^4 - 5 = 11$. Seven dimensions are inherited:

$$\theta \otimes \theta, \quad \theta \otimes u_i, \quad u_i \otimes \theta, \quad i = 0, 1, 2.$$

A convenient basis of the four-dimensional fresh sector K_2 is

$$\begin{aligned} \kappa_1 &= u_0 \otimes u_1 - u_1 \otimes u_0, \\ \kappa_2 &= u_0 \otimes u_2 - u_2 \otimes u_0, \\ \kappa_3 &= u_1 \otimes u_2 - u_2 \otimes u_1, \\ \kappa_4 &= u_0 \otimes u_2 + u_2 \otimes u_0 - 2u_1 \otimes u_1. \end{aligned}$$

Each vector has zero anti-diagonal sum under C_2 .

Corollary 8.2 (No globally one-dimensional phase model). *The phase direction 01 – 10 cannot close under block doubling. Any exact scale description retaining all ordering information must allow additional phase directions, including a fresh m^2 -dimensional sector at scale $m \rightarrow 2m$.*

This corollary is kinematic: it concerns the ambient phase space. The local fixed-block subset selected by positivity and causality will be characterized exactly in Section 14. The genuinely open object is smaller still: the subset whose descendants remain zero-repair viable across successive scales.

9 The cubic defect as a spectral normal form

Let E be the forward shift on test functions,

$$(Ef)(t) := f(t + 1).$$

For the dyadic pair define

$$P^* := \frac{1}{4}E^2 + \frac{3}{4}E^4, \quad Q^* := \frac{3}{4}E^3 + \frac{1}{4}E^5.$$

Theorem 9.1 (Exact cubic shift factorization).

$$\boxed{P^* - Q^* = \frac{1}{4}E^2(I - E)^3.} \tag{65}$$

Equivalently,

$$\boxed{\mu_P - \mu_Q = \frac{1}{4}\delta_2 * (\delta_0 - \delta_1)^{*3}.} \tag{66}$$

Proof. Expand $\frac{1}{4}E^2(I - E)^3$. The coefficients are $\frac{1}{4}(1, -3, 3, -1)$ on shifts 2, 3, 4, 5, exactly those of $P^* - Q^*$. $\square$

Thus every polynomial of degree at most two is annihilated. On the cubic,

$$(P^* - Q^*)t^3 = -\frac{3}{2}. \tag{67}$$

The first transverse polynomial jet is therefore one-dimensional modulo quadratics.

More generally, if a finite signed measure $\nu = \sum_k \nu_k \delta_k$ has vanishing raw moments through order $r - 1$, its generating polynomial $V(z) = \sum_k \nu_k z^k$ is divisible by $(1 - z)^r$. This is simply the invertible triangular relation between raw moments and falling-factorial moments. On test functions, the associated difference operator factors by $(I - E)^r$.

Laplace characters

$$f_a(t) := e^{-at}$$

are eigenfunctions of the shift, $Ef_a = e^{-a}f_a$. Therefore

$$(P^* - Q^*)f_a = \frac{1}{4}e^{-2a}(1 - e^{-a})^3 f_a. \quad (68)$$

The transform used in Section 3 is thus a spectral character of the exact translation algebra, not an unrelated trick.

Lemma 9.2 (Shift invariance of viability). *If $B \in \mathcal{K}_n$, then $z^s B \in \mathcal{K}_n$ for every integer $s \geq 0$.*

Proof. Multiply (16) at every node of a feasible tree by z^s . □

Remark 9.3 (Why Laplace modes are multipliers, not bill coordinates). If $a > 0$ and $\mathcal{K}_n \neq \emptyset$, then

$$\ell_n(e^{-a\cdot}) = 0.$$

Indeed, shift any fixed feasible law by s : its pairing with $e^{-a\cdot}$ is multiplied by e^{-as} and therefore tends to zero as $s \rightarrow \infty$; nonnegativity gives the reverse inequality. Laplace modes are therefore useful as dual multipliers and shift eigencharacters, not as direct coordinates of $\mathcal{B}_n = \ell_n(\text{id})$.

9.1 Why the third order is critical for this converse

Suppose more generally that

$$d(a) := -\log \rho(a) = c_r a^r + O(a^{r+1}), \quad c_r > 0,$$

while $\log \Lambda(a) = \sigma^2 a^2 + O(a^3)$. Then

$$\Gamma(a) = 2a + \frac{\sigma^2}{c_r} a^{3-r} (1 + o(1)). \quad (69)$$

For this proof mechanism, $r = 3$ is critical: $r = 2$ makes the logarithmic scale too small, $r = 3$ gives a finite nonzero coefficient, and $r > 3$ requires a different scaling or a sharper dual. This is a statement about the present converse mechanism, not a universal exponent theorem.

10 Green coordinates and boundary renormalization

Set

$$P(z) := 1 + 3z^2, \quad A(z) := z(3 + z^2).$$

Then

$$P(z) - A(z) = (1 - z)^3. \quad (70)$$

The original PGFs are regular at $z = 1$; the analytic difficulty is the triple zero. We resolve it by dividing by the corresponding Green kernel.

Definition 10.1 (Cubic Green potential). For a buffer PGF B , define

$$T_B(z) := \frac{1 - B(z)}{(1 - z)^3}. \quad (71)$$

Theorem 10.2 (Exact affine Green conjugacy). *The one-step zero-repair recursion (16) is equivalent to*

$$\boxed{T_{B_0}(z) + 3z^2 T_{B_1}(z) = A(z)T_B(z) + 1.} \quad (72)$$

Proof. Using $P - A = (1 - z)^3$,

$$\begin{aligned} (1 - B_0) + 3z^2(1 - B_1) &= P - AB \\ &= A(1 - B) + (P - A). \end{aligned}$$

Divide by $(1 - z)^3$. □

Thus the signed cubic source becomes the positive unit source. Positivity of the probability cone, rather than inversion of the cubic difference, is the remaining barrier.

Let $\beta_n := \mathbb{P}_B(J > n)$ and $T_B(z) = \sum_{n \geq 0} t_n z^n$. Direct coefficient extraction gives

$$t_n = \sum_{k=0}^n (n - k + 1) \beta_k. \quad (73)$$

With the backward difference $\nabla t_n = t_n - t_{n-1}$,

$$\boxed{\nabla t_n = \sum_{k=0}^n \beta_k, \quad \nabla^2 t_n = \beta_n, \quad \nabla^3 t_n = -\mathbb{P}(J = n) \ (n \geq 1).} \quad (74)$$

In particular,

$$\boxed{\mathbb{E}_B J = \sum_{n \geq 0} \beta_n = \lim_{n \rightarrow \infty} \nabla t_n.} \quad (75)$$

The bill is therefore an asymptotic slope in Green coordinates.

Iterating (72) for m generations gives

$$\sum_{w \in \{0,1\}^m} 3^{\text{wt}(w)} z^{2 \text{wt}(w)} T_{B_{hw}}(z) = A(z)^m T_{B_h}(z) + G_m(z), \quad (76)$$

where

$$\boxed{G_m(z) = \frac{P(z)^m - A(z)^m}{(1 - z)^3} = \sum_{j=0}^{m-1} P(z)^{m-1-j} A(z)^j.} \quad (77)$$

Hence G_m is coefficientwise nonnegative. At $m = 2$,

$$\boxed{G_2(z) = P(z) + A(z) = (1 + z)^3.} \quad (78)$$

The normalized mask $(1 + z)^3/4$ is the standard binary quadratic B-spline subdivision mask (B-spline order three). The connection is exact at the algebraic level: subdivision theory relates such masks, vanishing moments and polynomial reproduction [6].

Define the positive boundary dilation

$$\boxed{(\mathcal{R}F)(z) := \frac{1}{4}(1 + z)^3 F(z^2).} \quad (79)$$

This is not asserted to be the full nonlinear Bellman renormalization. It is the exact scale map in the cubic Green boundary algebra.

Proposition 10.3 (Exact third-difference conjugacy). *Let $\Delta_3 F := (1 - z)^3 F$ and $(D_2 G)(z) := G(z^2)$. Then*

$$\boxed{\Delta_3 \mathcal{R} = \frac{1}{4} D_2 \Delta_3.} \quad (80)$$

Proof. Use $(1 - z)^3(1 + z)^3 = (1 - z^2)^3$. □

Proposition 10.4 (Boundary pole spectrum). *If*

$$F(z) = \frac{c + o(1)}{(1 - z)^r} \quad (z \uparrow 1),$$

then

$$\boxed{(\mathcal{R}F)(z) = 2^{1-r} \frac{c + o(1)}{(1 - z)^r}.} \quad (81)$$

Thus pole orders $r = 0, 1, 2$ have multipliers $2, 1, 1/2$, respectively. The word “spectrum” here refers only to the exact boundary scaling law (81); no spectrum of the full viability operator has yet been proved.

11 An explicitly controllable marginal phase channel

A two-block causal correction arising in the dyadic model is the signed packet

$$\boxed{\eta(z) = \frac{1}{16}(z^2 + z^4 - 1 - z^6) = -\frac{1}{16}(1 - z^2)(1 - z^4).} \quad (82)$$

It is mass- and mean-neutral:

$$\eta(1) = 0, \quad \eta'(1) = 0,$$

while

$$\eta''(1) = -1. \quad (83)$$

Under the signed residual dilation

$$(\mathcal{S}\Theta)(z) := \frac{1}{4}\Theta(z^2),$$

the factorial moments $M_j = \Theta^{(j)}(1)$ transform as

$$\begin{aligned} M_0 &\mapsto \frac{1}{4}M_0, \\ M_1 &\mapsto \frac{1}{2}M_1, \\ M_2 &\mapsto M_2 + \frac{1}{2}M_1, \\ M_3 &\mapsto 2M_3 + 3M_2. \end{aligned}$$

Hence M_2 is exactly marginal on the subspace $M_0 = M_1 = 0$.

Apply the cubic Green transform to η :

$$\boxed{\Phi(z) := -\frac{\eta(z)}{(1 - z)^3} = \frac{(1 + z)^2(1 + z^2)}{16(1 - z)}.} \quad (84)$$

Its coefficients are

$$[z^n]\Phi = \frac{1}{16}(1, 3, 5, 7, 8, 8, 8, \dots),$$

so Φ lies in the simple-pole marginal sector of (81) and carries zero asymptotic mean slope.

Proposition 11.1 (Positive cohomological defect).

$$\boxed{\Phi - \mathcal{R}\Phi = \frac{(1+z)^3(1+z^2)(z^4+2z^2+3)}{64}.} \quad (85)$$

The right-hand side has nonnegative coefficients. Consequently,

$$0 \leq \mathcal{R}^k \Phi \leq \Phi \quad (k \geq 0) \quad (86)$$

coefficientwise.

Proof. Direct substitution into (79) gives (85). Positivity of $\mathcal{R}$ then allows iteration. $\square$

Define

$$\eta_s(z) := 4^{-s} \eta(z^{2^s}), \quad s \geq 0,$$

and

$$\boxed{E_*(z) := \frac{2}{3}z^2 + \sum_{s \geq 0} \frac{4^{-s}}{16} \left(1 + z^{2^{s+1}} + z^{2^{s+2}} + z^{6 \cdot 2^s}\right).} \quad (87)$$

Theorem 11.2 (Universal finite-mean envelope for the explicit phase channel). *The series E_* is a probability PGF with*

$$\boxed{\mathbb{E}_{E_*} J = \frac{17}{6}.} \quad (88)$$

For every sequence $(\theta_s)_{s \geq 0}$ with $|\theta_s| \leq 1$,

$$E_*(z) + \sum_{s \geq 0} \theta_s \eta_s(z) \quad (89)$$

is again a probability PGF with the same mean $17/6$.

Proof. The envelope part has total mass

$$\sum_{s \geq 0} \frac{4 \cdot 4^{-s}}{16} = \frac{1}{3},$$

which together with $2/3$ from the first atom gives mass one. The first term contributes mean $4/3$, while scale s contributes

$$\frac{4^{-s}}{16} \left(2^{s+1} + 2^{s+2} + 6 \cdot 2^s\right) = \frac{3}{4} 2^{-s}.$$

Summing yields $4/3 + 3/2 = 17/6$. Coefficientwise, E_* dominates the sum of absolute masses of every packet that can contribute at that coefficient; the triangle inequality gives nonnegativity of (89). Each η_s has zero mass and zero first moment, so normalization and mean are unchanged. $\square$

Remark 11.3 (The limit of this theorem). Theorem 11.2 controls one explicit phase channel through arbitrarily many scales at $O(1)$ mean cost. It does not show that the full causal phase fiber decomposes into copies of this channel. Theorem 8.1 in fact warns that fresh phase dimensions are created at every blocking step.

12 A critical self-similar reservoir

The dilation algebra also contains a canonical Mahler fixed point. Define

$$(\mathfrak{M}B)(z) := \frac{3}{4}z^2 + \frac{1}{4}B(z^2). \quad (90)$$

Proposition 12.1 (Mahler fixed point). *There is a unique probability PGF satisfying $B = \mathfrak{M}B$, namely*

$$B_*(z) = \sum_{k \geq 1} \frac{3}{4^k} z^{2^k}. \quad (91)$$

It has

$$\mathbb{E}_{B_*} J = 3, \quad \mathbb{E}_{B_*} J^2 = \infty. \quad (92)$$

Proof. Iterating $B(z) = \frac{3}{4}z^2 + \frac{1}{4}B(z^2)$ determines the coefficients and gives (91); its masses sum to one. The first moment is

$$\sum_{k \geq 1} \frac{3}{4^k} 2^k = 3,$$

while the second moment is

$$\sum_{k \geq 1} \frac{3}{4^k} 4^k = \infty.$$

□

The fixed point is therefore exactly critical between finite first and second moments. This fact should not be overinterpreted: a fixed point of the static scale map is not automatically an infinite zero-repair tree, and Theorem 3.5 forbids any such tree with finite root mean. The Mahler law is best viewed as a canonical boundary object of the scale algebra, not as a completed causal construction.

13 Positive fibers over scalar information

The quotient $V_m \rightarrow \mathbb{R}^{m+1}$ identifies exactly which linear information is lost when one passes from ordered histories to total information. Positivity can also be described exactly before causality is imposed.

For $0 \leq k \leq m$ recall the normalized Hamming-layer vector

$$u_k^{(m)} := \frac{1}{\binom{m}{k}} \sum_{|w|=k} e_w,$$

and define the canonical section

$$\sigma_m(r) := \sum_{k=0}^m r_k u_k^{(m)}, \quad r = (r_0, \dots, r_m) \in \Delta_m. \quad (93)$$

Here Δ_m denotes the probability simplex on the $m+1$ Hamming layers.

Theorem 13.1 (Exact positive phase fiber). *For $r \in \Delta_m$, the set of probability laws on ordered histories with scalar profile r is*

$$\Pi_m^{-1}(r) \cap \Delta(\{0, 1\}^m) = \sigma_m(r) + \mathbb{F}_m(r), \quad (94)$$

where

$$\boxed{F_m(r) = \left\{ \theta \in W_m : \theta_w \geq -\frac{r_{|w|}}{\binom{m}{|w|}} \text{ for every } w \right\}.} \quad (95)$$

Equivalently, the fiber is the direct product, over Hamming layers, of scaled simplices of total mass r_k . If every $r_k > 0$, its affine dimension is

$$\sum_{k=0}^m \left(\binom{m}{k} - 1 \right) = 2^m - (m+1).$$

Proof. If ν is a probability vector with $\Pi_m \nu = r$, then the total mass of ν on the layer $|w| = k$ is r_k . Since $\sigma_m(r)$ places the uniform mass $r_k / \binom{m}{k}$ on that layer, the difference $\theta = \nu - \sigma_m(r)$ has zero sum on each layer, hence $\theta \in \ker \Pi_m = W_m$. The condition $\nu_w \geq 0$ is exactly (95). The converse is immediate. The product-of-simplices statement follows layer by layer, and the dimension formula follows by summing the dimensions $\binom{m}{k} - 1$. $\square$

Theorem 13.1 is the exact analogue of a coefficientwise positivity test at the *ambient* history level. It is not yet the causal criterion we ultimately need. Prefix compatibility cuts a generally much smaller subset out of this product of simplices. The two-symbol theorem in Section 7 proves that this cut is already strict at the first nontrivial block length.

14 Exact finite-block causal phase fibers

At a fixed block length the causal restriction can be characterized without an ansatz. Let $\Omega_m := \{0, 1\}^m$ and

$$\mathcal{J}_m := \mathbb{R}^{\Omega_m \times \Omega_m} \cong V_m \otimes V_m.$$

Define the joint scalarization

$$\boxed{\Sigma_m := \Pi_m \otimes \Pi_m, \quad \Sigma_m e_{(x,w)} = e_{(\text{wt}(x), \text{wt}(w))}.} \quad (96)$$

Since Σ_m is surjective,

$$\boxed{\dim \ker \Sigma_m = 4^m - (m+1)^2.} \quad (97)$$

Thus even before causality is imposed, scalar joint profiles discard an exponentially growing family of ordering directions.

For a scalar joint profile $r = (r_{ij})_{0 \leq i,j \leq m}$, put

$$\boxed{\tau_m(r) := \sum_{i,j=0}^m r_{ij} u_i^{(m)} \otimes u_j^{(m)}.} \quad (98)$$

Then $\Sigma_m \tau_m(r) = r$.

Let $\pi(x, w)$ be a nonnegative coupling of $P^{\otimes m}$ and $Q^{\otimes m}$. Besides the two marginal families, causality from W to X is equivalent to

$$\boxed{\sum_{x' \in \Omega_{m-t}} \pi((a, x'), (b, c)) = Q^{\otimes(m-t)}(c) \sum_{x' \in \Omega_{m-t}} \sum_{c' \in \Omega_{m-t}} \pi((a, x'), (b, c'))} \quad (99)$$

for $1 \leq t < m$, $a, b \in \Omega_t$, and $c \in \Omega_{m-t}$. These equations say exactly that, conditional on $W_{1:t} = b$, the law of $X_{1:t}$ is independent of the future $W_{t+1:m}$. Write all marginal and causal equations as

$$A_m \pi = b_m. \quad (100)$$

Redundant rows are allowed.

Definition 14.1 (Local causal phase fiber). For a scalar joint profile r , define

$$\boxed{C_m^{\text{loc}}(r) := \{\theta \in \ker \Sigma_m : \tau_m(r) + \theta \geq 0, A_m(\tau_m(r) + \theta) = b_m\}.} \quad (101)$$

Theorem 14.2 (Exact finite-block causal phase-fiber criterion). *For every m and every scalar joint profile r ,*

$$\boxed{C_m^{\text{loc}}(r) = \{\theta \in \ker \Sigma_m : A_m\theta = b_m - A_m\tau_m(r), \theta \geq -\tau_m(r)\}.} \quad (102)$$

Hence the local causal phase fiber is a bounded polytope, possibly empty. If P, Q, r are rational, it is a rational polytope.

Moreover, the fiber is nonempty if and only if

$$\boxed{\langle y, b_m \rangle + \langle z, r \rangle \geq 0} \quad (103)$$

for every pair (y, z) satisfying

$$\boxed{A_m^T y + \Sigma_m^T z \geq 0} \quad (104)$$

componentwise. For rational P, Q , the cone in (104) is rational polyhedral, hence a finite rational generating family of rays suffices.

Proof. Every coupling with scalar profile r has the unique form $\pi = \tau_m(r) + \theta$ with $\theta \in \ker \Sigma_m$. Substitution gives (102); boundedness follows because the translated fiber lies in the probability simplex.

For nonemptiness apply Farkas' lemma to

$$\pi \geq 0, \quad A_m\pi = b_m, \quad \Sigma_m\pi = r.$$

Such a nonnegative solution exists if and only if every (y, z) with $A_m^T y + \Sigma_m^T z \geq 0$ satisfies $\langle y, b_m \rangle + \langle z, r \rangle \geq 0$. Rational polyhedrality is immediate when the source probabilities are rational. $\square$

Remark 14.3 (The real remaining difficulty). Theorem 14.2 completely settles the *local fixed-block* causal fiber. It does not settle zero repair. A zero-repair block must land in child laws that remain viable for all remaining depths. Thus the hard question is no longer whether a finite inequality description exists at fixed m , but whether those certificates admit a scale-stable intrinsic recursion with uniform bill control under $m \mapsto 2m$.

Corollary 14.4 (Finite truncations admit exact certificates). *Every finite-horizon zero-repair problem with a finite support cutoff is a finite linear feasibility problem. Any projected coarse/phase fiber is therefore a rational polytope and admits a finite exact Farkas certificate family. Numerical LP duals can consequently be rationalized into exact certificates; what is not automatic is a support-independent, scale-uniform formula.*

15 Why finite closures cannot be the answer

The phase-space growth above might tempt one to search for a fortunate finite collection of exact buffer profiles or for one scalar potential that absorbs all of the ordering information. Both possibilities are ruled out.

Theorem 15.1 (No finite exact profile automaton). *There do not exist finitely many probability PGFs $B^{(1)}, \dots, B^{(K)}$ and transition maps $\tau_0, \tau_1 : \{1, \dots, K\} \rightarrow \{1, \dots, K\}$ such that*

$$z(3 + z^2)B^{(s)}(z) = B^{(\tau_0(s))}(z) + 3z^2B^{(\tau_1(s))}(z) \quad (105)$$

for every s .

Proof. Let $G = (B^{(1)}, \dots, B^{(K)})^\top$, and let L, H be the selector matrices induced by τ_0, τ_1 . Then

$$M(z)G(z) = 0, \quad M(z) := z(3 + z^2)I - L - 3z^2H.$$

The coefficient of z^{3K} in $\det M(z)$ is 1: it is obtained uniquely by selecting the z^3 term from every diagonal entry. Thus $\det M$ is not the zero polynomial. For all but finitely many $z \in (0, 1)$, $M(z)$ is invertible and therefore $G(z) = 0$, contradicting positivity of probability PGFs on $(0, 1)$. $\square$

Theorem 15.2 (No positive scalar state-only drift). *Suppose $\phi : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ and constants c, λ_x, μ_y satisfy*

$$\phi(j + y - x) - \phi(j) \geq c + \lambda_x + \mu_y \quad (106)$$

whenever $j + y - x \geq 0$, with gauges

$$\sum_x P(x)\lambda_x = 0, \quad \sum_y Q(y)\mu_y = 0.$$

For the dyadic pair, $c \leq 0$.

Proof. For the dyadic pair the difference $d := y - x$ takes values $-1, 1, 3$. The $+1$ inequalities give a uniform lower bound on $d_j := \phi(j + 1) - \phi(j)$, while the -1 inequality gives a uniform upper bound. Hence (d_j) is bounded.

Set $a_N = N^{-1} \sum_{j=1}^N d_j$. Along a subsequence $N_k \rightarrow \infty$, $a_{N_k} \rightarrow a$. For each fixed $r \in \{-1, 1, 3\}$, boundedness implies

$$\frac{1}{N_k} \sum_{j=1}^{N_k} [\phi(j + r) - \phi(j)] \rightarrow ra. \quad (107)$$

For positive r , expand into consecutive first differences; for $r = -1$, shift the index. The finitely many endpoint terms are negligible because (d_j) is bounded.

Average (106) over $j = 1, \dots, N_k$ and over $(X, Y) \sim P \otimes Q$. All moves are admissible for $j \geq 1$, and the gauge terms vanish. Passing to the subsequential limit using (107) gives

$$a \mathbb{E}(Y - X) \geq c.$$

The two increment laws have equal mean, so $\mathbb{E}(Y - X) = 0$, and therefore $c \leq 0$. $\square$

These theorems have a useful positive interpretation. A sharp asymptotic object, if it exists, must retain information at growing scales. The large kernel W_m is not an artifact of a poor coordinate system that can be compressed into finitely many exact modes without loss.

16 The renormalized positive phase problem

We can now state the unresolved problem without appealing to a particular buffer ansatz.

The linear quotient $V_m = U_m \oplus W_m$ separates visible total-information profiles from ordering information. Theorem 13.1 describes the complete positive fiber before causality, while Theorem 14.2 gives a complete finite-dimensional dual criterion for the local causal matching fiber at every fixed block length. Thus the unresolved issue is now more specific: *future zero-repair viability must be incorporated in a way that is stable under scale.*

The role of Section 8 is therefore decisive. Blocking from m to $2m$ does not merely transport an old phase vector. It creates the fresh sector K_m of dimension m^2 . Thus any exact scale map must account for at least three operations:

transport inherited phase + create inter-block phase + project back into the positive causal fiber.
--

The last operation is nonlinear because positivity and future viability are inequalities.

16.1 The central characterization problem

At fixed m , the local coupling fiber is no longer mysterious: Theorem 14.2 characterizes it exactly, and a finite set of rational dual rays exists in principle. What is missing is the analogue of a single structural criterion whose form survives scale doubling and already incorporates descendant viability.

Open Problem 16.1 (Scale-stable zero-repair phase characterization). *Find coarse/phase coordinates and a family of necessary-and-sufficient positivity or dual inequalities such that: (i) one-scale lifts land in child states viable for the prescribed remaining horizon; (ii) the form closes under $m \mapsto 2m$; and (iii) the induced repair cost is uniformly bounded in the dyadic scale. Equivalently, compress the exponentially expanding fixed-block Farkas descriptions into an intrinsic renormalization law on the positive zero-repair phase state.*

For $m = 2$, Proposition 6.3 isolates the unique one-sided phase coordinate Θ . For general m , Theorem 8.1 shows why a single Θ cannot suffice, and Theorem 14.2 shows that the obstacle is not finite-dimensional duality itself but the absence of a scale-uniform compression.

16.2 A conditional scale theorem

The following simple implication isolates exactly what an upper-bound theorem would have to deliver.

Proposition 16.2 (Bounded scale repair implies logarithmic growth). *Suppose there is a constant $C < \infty$ such that*

$$\boxed{\mathcal{B}_{2n} \leq \mathcal{B}_n + C} \tag{108}$$

for every $n \geq 1$. Then

$$\mathcal{B}_n = O(\log n). \tag{109}$$

For the dyadic parity pair, this combines with Theorem 4.1 to give

$$\boxed{\mathcal{B}_n = \Theta(\log n)}. \tag{110}$$

Proof. Iterating (108) gives $\mathcal{B}_{2^k} \leq \mathcal{B}_1 + kC$. Monotonicity of the finite-horizon viability bill then gives the same order for arbitrary n by choosing $2^{k-1} < n \leq 2^k$. The lower bound is Theorem 4.1. $\square$

Equation (108) is not proved here. It is the most concrete sharp upper target suggested by the scale algebra. If the dyadic increment converged,

$$\mathcal{B}_{2n} - \mathcal{B}_n \longrightarrow \gamma,$$

then along dyadic horizons one would obtain

$$\mathcal{B}_n \sim \frac{\gamma}{\log 2} \log n,$$

and the rigorous lower bound would force

$$\gamma \geq \frac{\log 2}{3}. \tag{111}$$

No equality is asserted.

16.3 Why the one-channel Jordan picture is insufficient

The explicit packet η_s is marginal in the second factorial moment, but its first-moment carrying cost decays geometrically: an envelope mass of order 4^{-s} transported to distance 2^s costs order 2^{-s} . This is exactly why Theorem 11.2 has finite mean. Consequently, one copy of the explicit η channel cannot by itself explain logarithmic growth.

A more plausible critical mechanism, if the logarithmic lower bound is sharp, is the repeated creation of fresh phase directions under blocking. The ambient fresh sector has dimension m^2 , while the cubic signed residual is suppressed by a third-order scale factor. The dimensional balance

$$m^2 \cdot m^{-3} \cdot m \asymp 1 \quad (112)$$

is suggestive: multiplicity, cubic residual mass, and first-moment transport distance cancel at order three. We emphasize that (112) is a scaling heuristic, not a theorem. It does not show that $\Theta(m^2)$ feasible phase directions are independently excited, nor that positivity prevents cancellations. Its value is that it identifies a falsifiable mechanism compatible with both the exact cubic symbol and the exact growth of the phase kernel.

17 Relation to neighboring mathematical theories

Adapted and causal optimal transport. The causal constraint in Section 7 is a standard nonanticipativity constraint on a coupling. General causal and bicausal transport duality, including dual attainment under broad measurable-cost hypotheses, is available in [9]. Very recent graph-causal optimal transport characterizes gluing properties and dynamic programming for causality prescribed by directed acyclic graphs [16]. These results make adapted transport a natural surrounding language. They do not make the present problem automatic: exact inventory conservation and the hard cone constraint $J \geq 0$ must still be incorporated, and the full countable-state no-gap theorem for our Bellman–Fenchel recursion has not been proved.

Convex dynamic programming. The passage from convex viability sets to lower support functionals is a finite-dimensional convex duality calculation, not a new general duality theorem. For infinite-dimensional stochastic optimization with information constraints, the recent work of Pennanen and Perkkiö provides a broader convex-duality and dynamic-programming framework [11, 12]. The unresolved issue here is to choose a weighted function/measure topology in which the unbounded mean objective, shift invariance and positivity coexist with strong duality.

Nonlinear Perron–Frobenius theory. Order-preserving homogeneous and additively homogeneous maps, quotienting by constants, and projective metrics are standard in nonlinear Perron–Frobenius theory and in Shapley operators [5, 8, 7]. Recent work also extends joint spectral-radius ideas to switched nonlinear order-preserving systems [10]. The relevance here is structural, not a theorem transfer: Theorem 15.1 already rules out an exact finite-mode state. Across all scales the zero-repair phase state is necessarily infinite-dimensional and constrained by positivity.

Subdivision and vanishing moments. The identity

$$\Delta_3 \mathcal{R} = \frac{1}{4} D_2 \Delta_3$$

places the boundary algebra directly next to subdivision schemes. The mask $(1+z)^3/4$ is the binary quadratic B-spline mask; its order-three zero condition is the same algebra that annihilates the matched mass, mean and variance jets. Sum rules, polynomial reproduction and invariant

difference spaces are classical tools in subdivision theory [6]. The novelty of the present difficulty is the additional causal-positive fiber sitting over this linear scheme.

Finitary coding and randomness recycling. The zero-repair equation resembles a constrained information coboundary, while finitary isomorphism theory shows that stronger moment assumptions on coding length preserve increasingly fine information invariants [2, 3, 4]. In a different online-sampling model, modern randomness-recycling algorithms can approach the Shannon limit using persistent state; recent real-probability algorithms achieve expected entropy $\sum_i H(F_i) + O(\log n)$ with $O(\log n)$ persistent space [13, 14]. Neither result implies a zero-repair theorem, but both reinforce the need to distinguish entropy rate from persistent working state.

Synchronization-channel duality. Kovačević’s recent exact sticky-channel calculation is a useful nearby example of proof architecture rather than a source of a theorem used here: a channel is reduced to a simpler operational object, a dual bound is made exact by a coefficientwise criterion, and the entire domination class is then characterized algebraically [15]. Problem 16.1 asks for the analogous structural step for a causal-positive phase fiber.

18 Back to “Who pays the bill?”

The original resource question was broader than the zero-repair model. An observable stochastic system may be realizable at the ordinary Shannon rate while different exact-realization contracts impose different requirements on its hidden state. The mathematics above separates three quantities that should not be conflated:

$$\boxed{\text{entropy rate} \neq \text{fresh repair expenditure} \neq \text{exact causal working inventory.}} \quad (113)$$

The first is a rate: how many new random bits per step are required asymptotically. The second measures information that a simulator is permitted to inject after seeing where mismatch occurred. The third is the present quantity: a nonnegative inventory that must make every one-step information identity exact without such fresh repair.

For the dyadic parity pair, the present paper proves a sharp qualitative statement about the third resource. Even though mass, mean and variance of the two information laws agree, their third-order mismatch cannot be carried forever by any finite-mean zero-repair inventory:

$$\mathcal{B}_n \geq \left(\frac{1}{3} - o(1) \right) \log n. \quad (114)$$

At the same time, the two-symbol causal matching theorem shows why a scalar information spectrum alone is insufficient: causality creates an additional positive obstruction while leaving the signed scalar residual unchanged. Thus part of the bill is carried by ordering information that disappears under the usual scalar projection.

This does *not* prove that an unrestricted causal simulator pays an additional asymptotic entropy rate. In the broader causal-tree model, fresh randomness, nonlocal blocks and hidden dependence may implement strategies forbidden by zero repair. A companion analysis of that broader contract finds that, for two finite actions, the first-order causal rate can reach the observable Shannon floor [17]. There is no contradiction: rate and working inventory are different resources.

The missing bridge is therefore mathematically precise.

Open Problem 18.1 (Entropy–inventory comparison). *Establish the sharpest general comparison, if any, between the finite-horizon unrestricted causal entropy residual and the zero-repair viability bill $\mathcal{B}_n$. In particular, determine which forms of fresh repair or blockwise hidden dependence can convert a diverging working-inventory requirement into only $o(n)$ additional source entropy.*

The answer would return the phase-renormalization theory developed here to the original accounting problem. Until such a bridge is proved, the correct conclusion is operational rather than ontological: exact realization costs depend on what must remain simultaneously valid, on what may be repaired later, and on which history information the hidden state is allowed to retain.

19 Theorem ledger and open frontier

For clarity, Table 1 separates the proved statements from the renormalization program.

Statement	Status
Finite-horizon fixed-tilt logarithmic converse	proved
Third-cumulant lower bound $\liminf \mathcal{B}_n / \log n \geq \Delta\kappa_3 / (6\sigma^2)$	proved
Dyadic lower coefficient $1/3$ and $\mathcal{B}_n \rightarrow \infty$	proved
Finite-dimensional Bellman–Fenchel duality	proved
Countable-state Bellman formula as weak dual certificate	proved
Phase quotient $W_m = \ker \Pi_m$, $\dim W_m = 2^m - (m + 1)$	proved
Exact positive scalar-profile fiber, Theorem 13.1	proved
Fixed-block local causal phase fiber: exact polyhedral/Farkas characterization	proved
Two-symbol projected causal profile: exact two-facet characterization	proved
Two-symbol causal mismatch $5/16$ with solver-free proof and exact primal–dual certificate	proved
Causal surcharge invisible to the signed scalar residual	proved
Block-doubling decomposition of W_{2m} and fresh sector $\dim K_m = m^2$	proved
Cubic shift factorization and Green affine conjugacy	proved
Explicit marginal phase cohomology and mean- $17/6$ envelope	proved*
Mahler scale fixed point with finite first and infinite second moment	proved
Finite exact profile automaton	impossible
Positive scalar state-only Bellman drift	impossible
Scale-stable characterization of the zero-repair causal phase state	open
Strong countable-state Bellman–Fenchel duality in a natural weighted space	open
Uniform doubling bound $\sup_n (\mathcal{B}_{2n} - \mathcal{B}_n) < \infty$	open
Matching $O(\log n)$ upper bound / $\Theta(\log n)$ law	open
Exact asymptotic constant for $\mathcal{B}_n / \log n$	open
General comparison with unrestricted causal entropy residual	open

Table 1: Logical status of the main statements. The symbol * means: proved for the explicit phase channel stated in Section 11, not for the full zero-repair phase dynamics.

The frontier can be summarized in one sentence:

the cubic linear defect and every fixed-block causal fiber are explicit;
the remaining object is their scale-stable zero-repair renormalization.

20 Conclusion

The first nontrivial dyadic zero-repair problem contains two distinct kinds of information. The scalar information profile records how much surprisal has accumulated; the phase fiber records in which order that information was created. Static convolution sees only the first. Causality sees both.

This distinction is already completely visible after two symbols. The projection of the causal coupling polytope is the static scalar transport polytope cut by two explicit oblique facets. One of these facets alone forces the static mismatch $1/4$ up to the exact causal value $5/16$. The additional $1/16$ appears identically on both scalar residuals and is therefore invisible to their

signed difference. At larger block lengths the hidden space expands rapidly: W_m has dimension $2^m - (m + 1)$, and every doubling creates a fresh m^2 -dimensional anti-diagonal sector even before positivity is imposed.

Independently, the source mismatch has an exact cubic normal form. The factor $(I - E)^3$ annihilates the matched mass, mean and variance jets; division by the corresponding cubic zero produces a Green coordinate with a positive unit source and a quadratic B-spline boundary mask. The same order three is critical in the rigorous Laplace converse, which forces

$$\mathcal{B}_n \geq \left(\frac{1}{3} - o(1)\right) \log n$$

for the dyadic pair.

The two structures meet at the unresolved point. Linear cubic inversion and fixed-block causal feasibility are explicit; keeping all descendants inside a positive, prefix-compatible *future-viable* phase state is not. A scale-stable characterization of that state would turn the present renormalization picture into an actual upper-bound mechanism. A bounded cost per dyadic scale would close the growth order at $\Theta(\log n)$.

The broader information-theoretic lesson is narrower but already rigorous: an exact stochastic realization can have different costs depending on whether one prices source entropy, fresh repair, or nonnegative causal working inventory. The scalar information spectrum does not contain all of the causal bookkeeping. Ordering information can carry a bill of its own.

A Complete rational certificate for the two-symbol causal LP

This appendix gives an independent full rational LP certificate for Corollary 7.3. The closed-form two-facet proof in Section 7 does not require this appendix, but the certificate makes the original sixteen-variable formulation fully checkable without a solver. Vectorize the coupling matrix row by row as

$$\pi = (\pi_{00,00}, \pi_{00,01}, \dots, \pi_{11,11})^T.$$

With the constraint order used in the proof, the equality matrix is

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ \frac{1}{4} & -\frac{3}{4} & 0 & 0 & \frac{1}{4} & -\frac{3}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{4} & \frac{3}{4} & 0 & 0 & -\frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & -\frac{3}{4} & 0 & 0 & \frac{1}{4} & -\frac{3}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{4} & \frac{3}{4} & 0 & 0 & -\frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & -\frac{3}{4} & 0 & 0 & \frac{1}{4} & -\frac{3}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & \frac{3}{4} & 0 & 0 & -\frac{1}{4} & \frac{3}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & -\frac{3}{4} & 0 & 0 & \frac{1}{4} & -\frac{3}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & \frac{3}{4} & 0 & 0 & -\frac{1}{4} & \frac{3}{4} \end{pmatrix}. \quad (115)$$

The right-hand side is (54). The first eight rows impose the two marginals; the last eight are the causal equations (35). Redundant rows are retained to match the lexicographic construction.

For the dual vector (55), direct multiplication gives

$$A^T y = \begin{pmatrix} 1 & 1 & 1 & -1 \\ 0 & 0 & 0 & -2 \\ 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Thus $A^T y \leq C$ componentwise. Finally,

$$b^T y = \frac{1}{16} \left(1 - \frac{3}{4} + \frac{27}{4} - 2 \right) = \frac{5}{16}.$$

Together with the primal matrix (53), this is an exact rational primal–dual certificate of the optimum.

Research-status and reproducibility note

All asymptotic claims in this paper are supported by analytic arguments. The exact two-symbol causal value is accompanied by explicit rational primal and dual certificates, so it does not depend on floating-point optimization. Computer algebra was used to check finite polynomial identities and exact coefficients during development, not as a substitute for the stated proofs.

The paper deliberately uses conservative priority language. The surrounding subjects—adapted transport, finitary coding, nonlinear positive operators, subdivision theory and synchronization channels—are mature, and an equivalent formulation of some structural lemma may exist elsewhere. Where a direct precedent is known, it is cited; no stronger historical novelty claim is intended.

During preparation, the author used OpenAI’s ChatGPT as a research assistant for literature discovery, algebraic checking, adversarial proof review and L^AT_EX preparation. The mathematical statements are separated explicitly into proved results and open problems in Table 1.

References

- [1] R. T. Rockafellar, *Convex Analysis*, Princeton University Press, 1970.
- [2] W. Parry, *Finitary isomorphisms with finite expected code lengths*, Bull. London Math. Soc. **11** (1979), 170–176. <https://doi.org/10.1112/blms/11.2.170>
- [3] K. Schmidt, *Invariants for finitary isomorphisms with finite expected code lengths*, Invent. Math. **76** (1984), 33–40.
- [4] N. Harvey and Y. Peres, *An invariant of finitary codes with finite expected square root coding length*, Ergodic Theory Dynam. Systems **31** (2011), 77–90. <https://doi.org/10.1017/S014338570900090X>
- [5] B. Lemmens and R. D. Nussbaum, *Nonlinear Perron–Frobenius Theory*, Cambridge Tracts in Mathematics, Vol. 189, Cambridge University Press, 2012.
- [6] M. Charina and C. Conti, *Polynomial reproduction of multivariate scalar subdivision schemes*, J. Comput. Appl. Math. **240** (2013), 51–61. <https://doi.org/10.1016/j.cam.2012.06.013>
- [7] M. Akian, S. Gaubert, and A. Hochart, *A game theory approach to the existence and uniqueness of nonlinear Perron–Frobenius eigenvectors*, Discrete Contin. Dyn. Syst. **40** (2020), 207–231. <https://doi.org/10.3934/dcds.2020009>

- [8] B. Lins, *A unified approach to nonlinear Perron–Frobenius theory*, Linear Algebra Appl. **675** (2023), 48–89. <https://doi.org/10.1016/j.laa.2023.06.018>
- [9] D. Kršek and G. Pammer, *General duality and dual attainment for adapted transport*, Appl. Math. Optim. **91** (2025), Article 52. <https://doi.org/10.1007/s00245-025-10240-y>
- [10] P. Deidda, N. Guglielmi, and F. Tudisco, *Nonlinear joint spectral radius*, arXiv:2507.11314, 2025.
- [11] T. Pennanen and A.-P. Perkkiö, *Duality in convex stochastic optimization*, Math. Program. **215** (2026), 145–191. <https://doi.org/10.1007/s10107-025-02216-1>
- [12] T. Pennanen and A.-P. Perkkiö, *Dynamic programming and dimensionality in convex stochastic optimization and control*, Optim. Lett. **20** (2026), 257–273. <https://doi.org/10.1007/s11590-025-02230-4>
- [13] T. L. Draper and F. A. Saad, *Efficient online random sampling via randomness recycling*, in Proc. 2026 ACM–SIAM Symposium on Discrete Algorithms (SODA), 2026, pp. 2473–2511. <https://doi.org/10.1137/1.9781611978971.89>
- [14] T. L. Draper, D. G. Harris, and F. A. Saad, *Online random sampling with real probabilities*, arXiv:2607.13828, 2026.
- [15] M. Kovačević, *The capacity of a family of sticky channels*, arXiv:2607.28281, 2026.
- [16] J. Oblój and V. Tuchilus, *Graph causal optimal transport and Wasserstein distances*, arXiv:2608.13716, 2026.
- [17] J. Mikulík, *Who Pays the Bill? Counterfactual Identity, the Causal Shannon Floor, Spectral Ledgers, and Orbit Resolution*, research preprint, 2026.